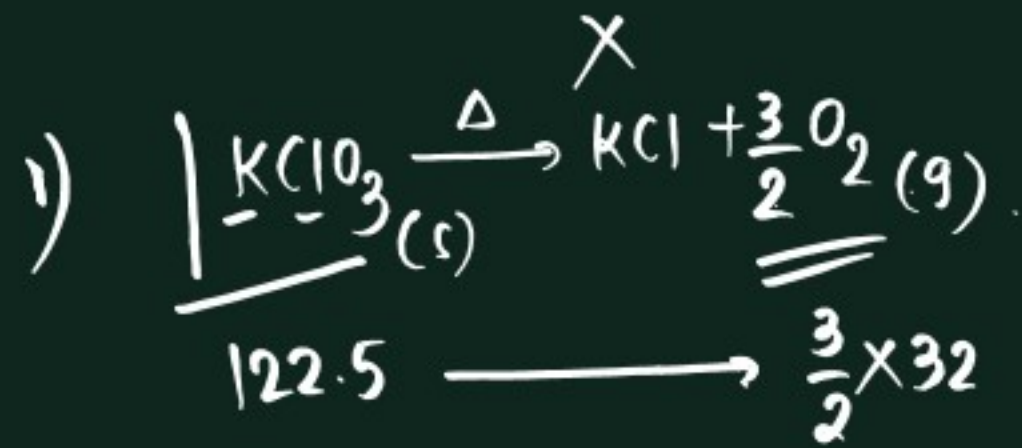




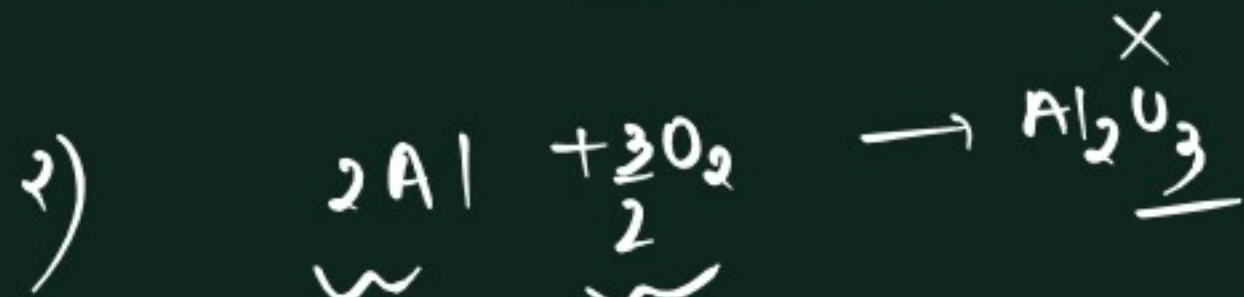
$$122.5 \longrightarrow \frac{3}{2} \times 32$$

$$122.5 \text{ gm} \longrightarrow 48$$

$$x \longleftarrow 24$$

$$x \times 48 = 24 \times 122.5$$

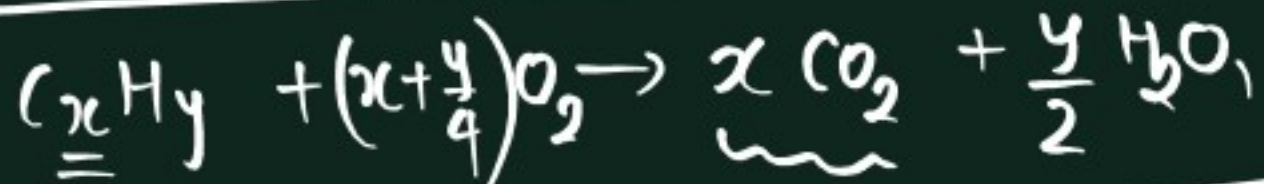
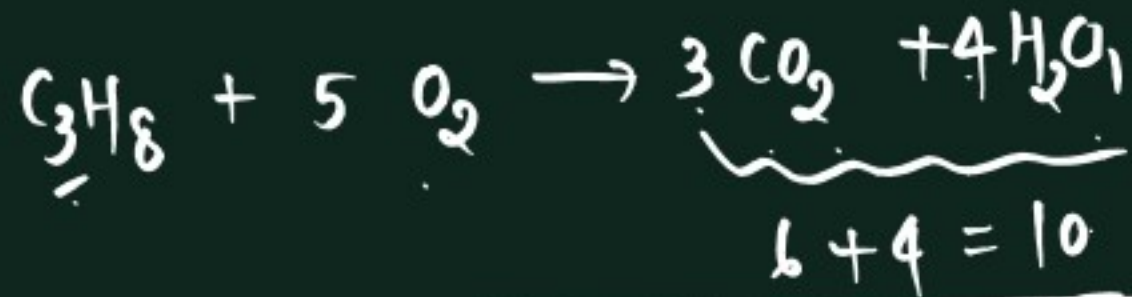
$$\underline{x = 6.125 \text{ gm}}$$



$$54 \longrightarrow 48$$

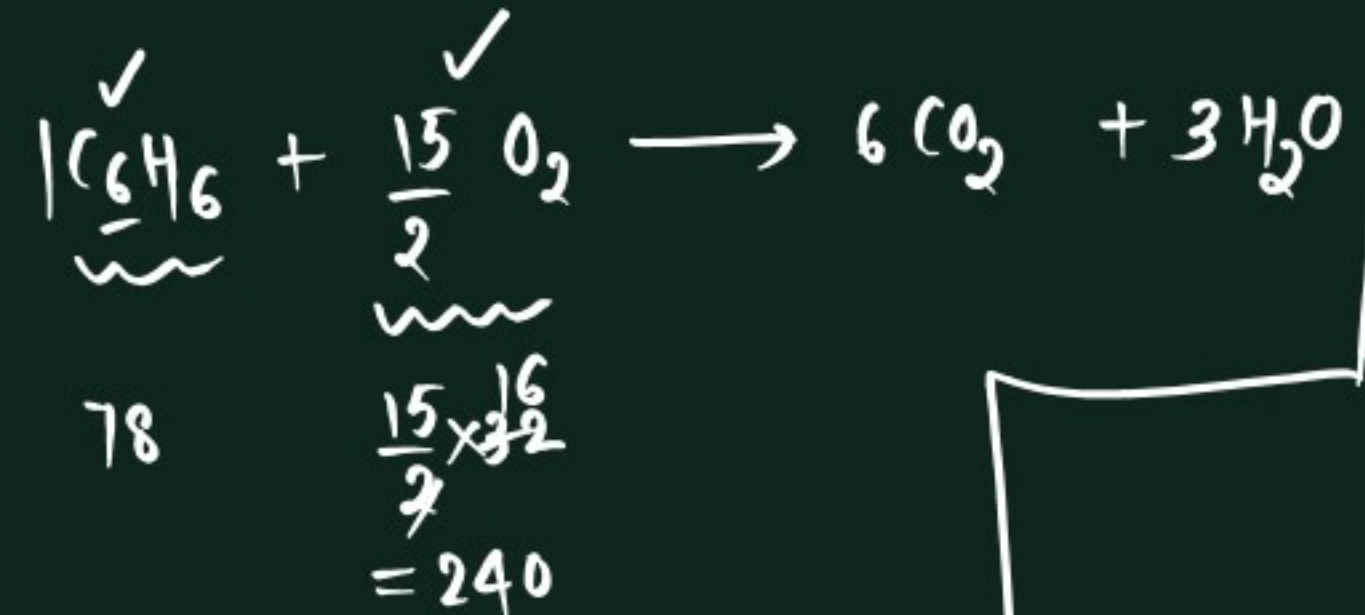
$$27 \longrightarrow x = 24 \text{ gm}$$

* Combustion reaⁿ $\rightarrow$ burning of Hydrocarbon

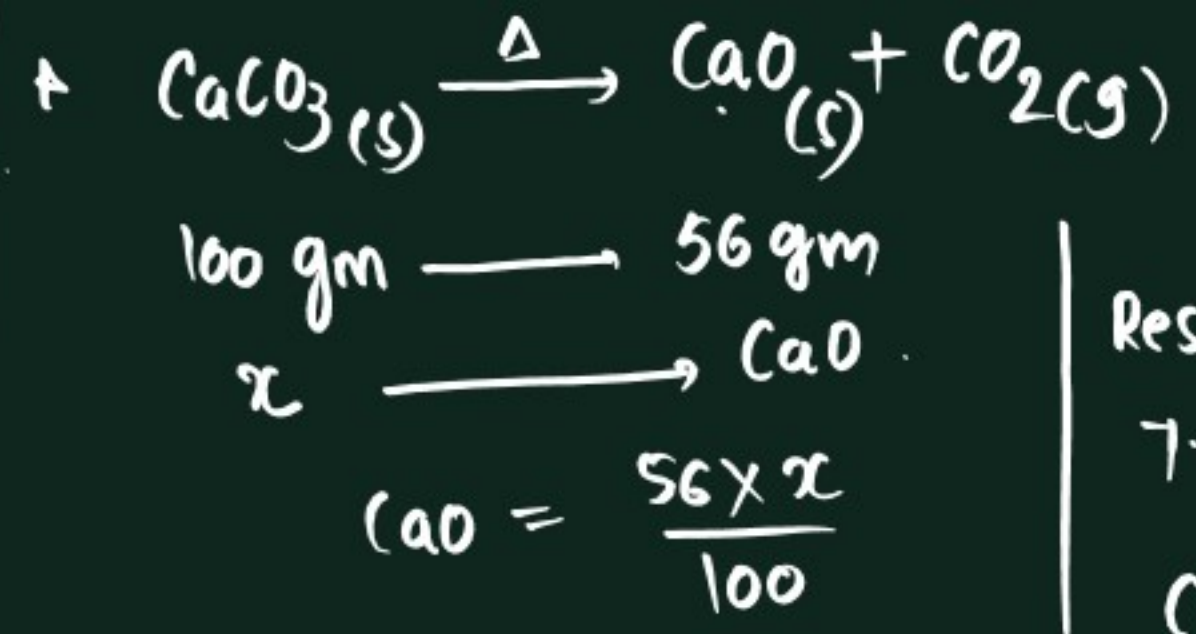
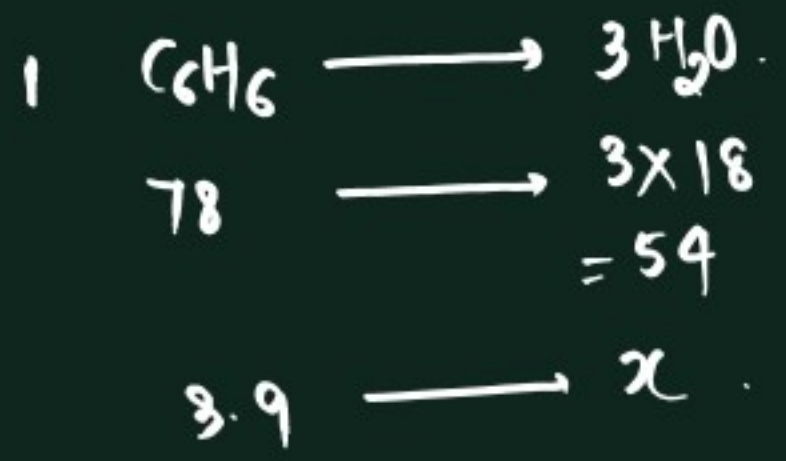


$$\frac{2x + \frac{y}{2}}{2}$$

$$\frac{30}{2}$$

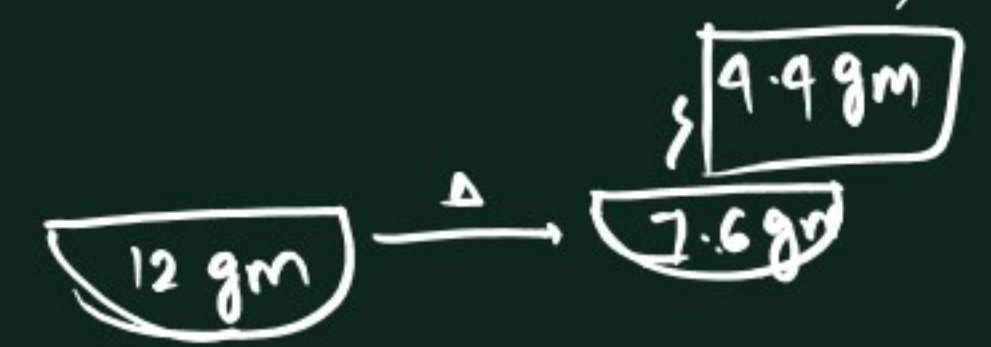


3.9 $\longrightarrow$ x
 $x = 12 \text{ gm}$



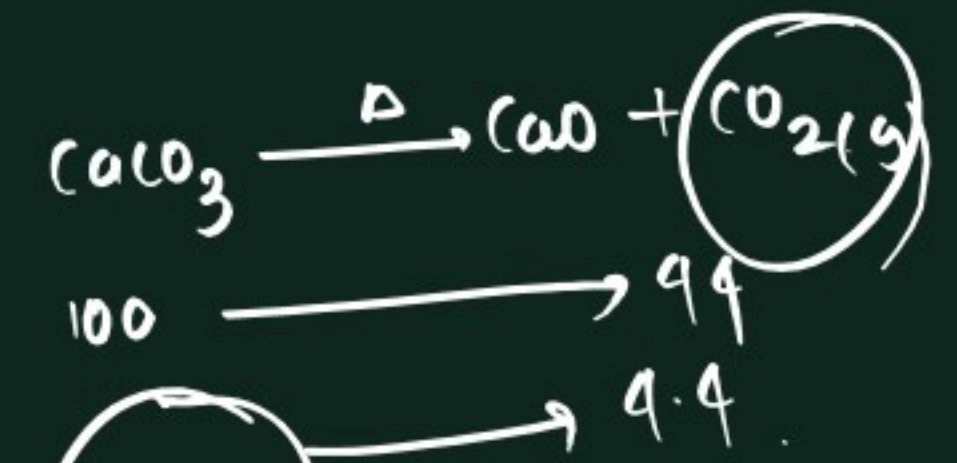
$CaCO_3 + \text{sand} = 12 \text{ gm}$
 $\underline{x} \quad \underline{(12-x)}$

Residue = CaO + sand.
 7.6 gm =
 $CaO = 7.6 - \text{sand}$
 $= 7.6 - (12 - x)$



$CO_2 = 4.4 \text{ gm}$

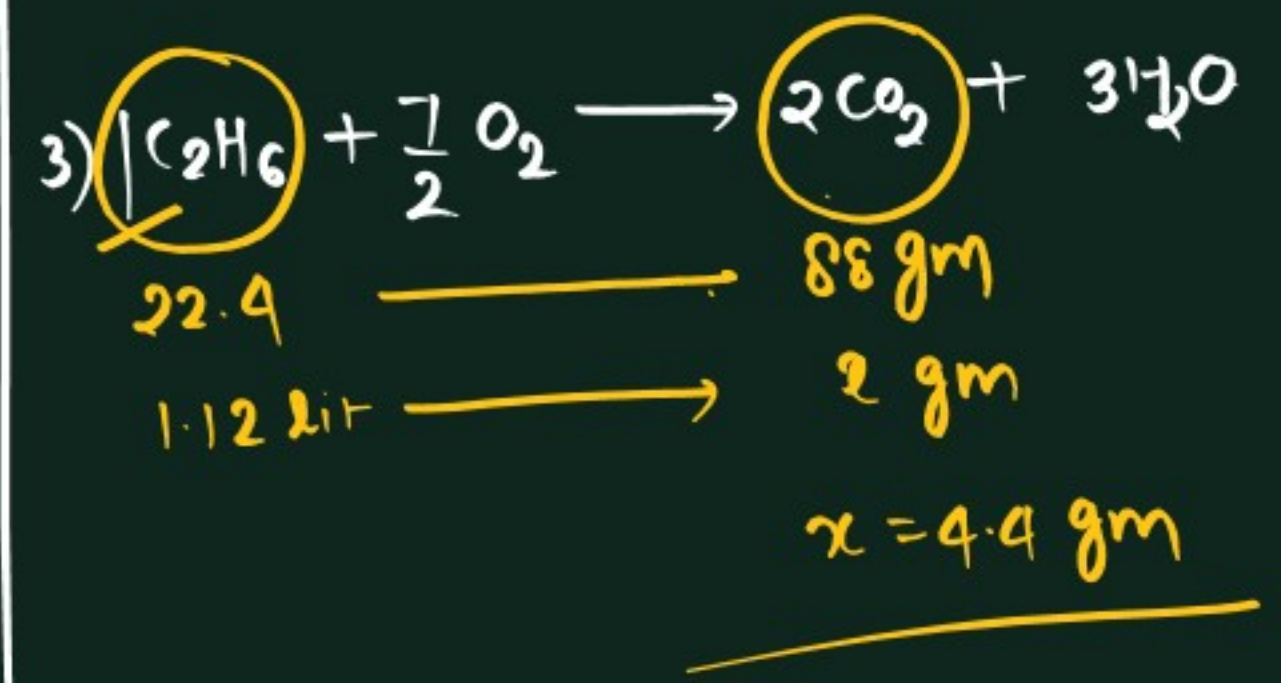
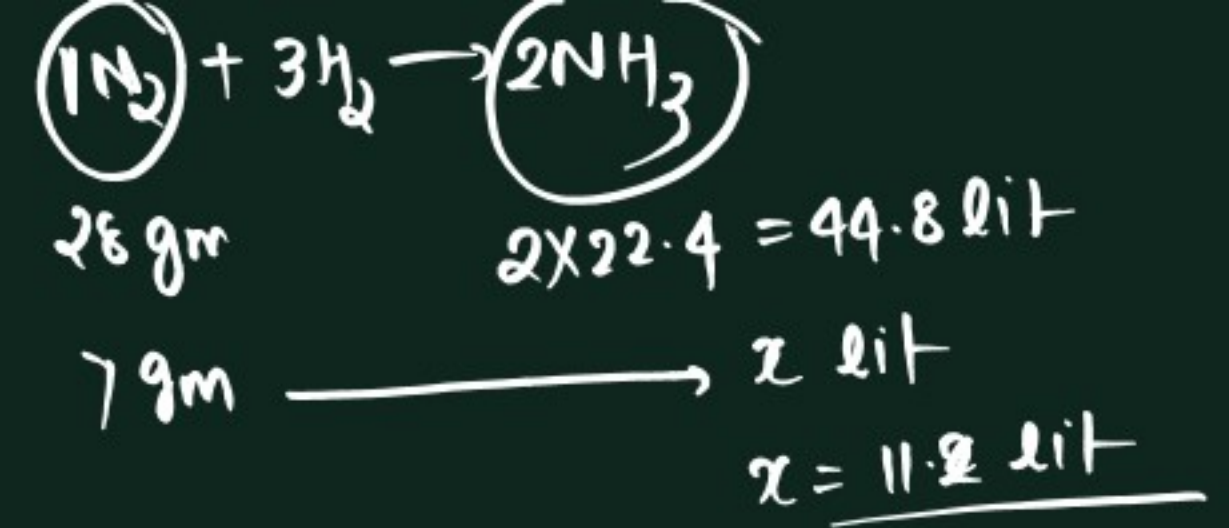
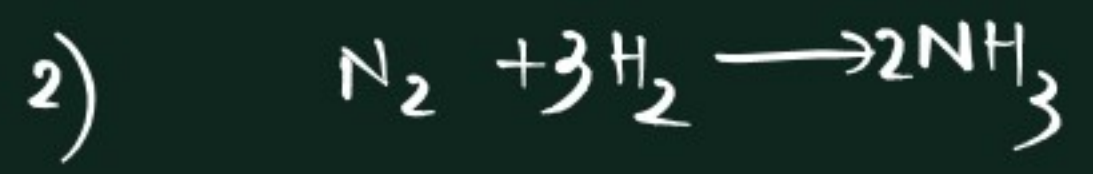
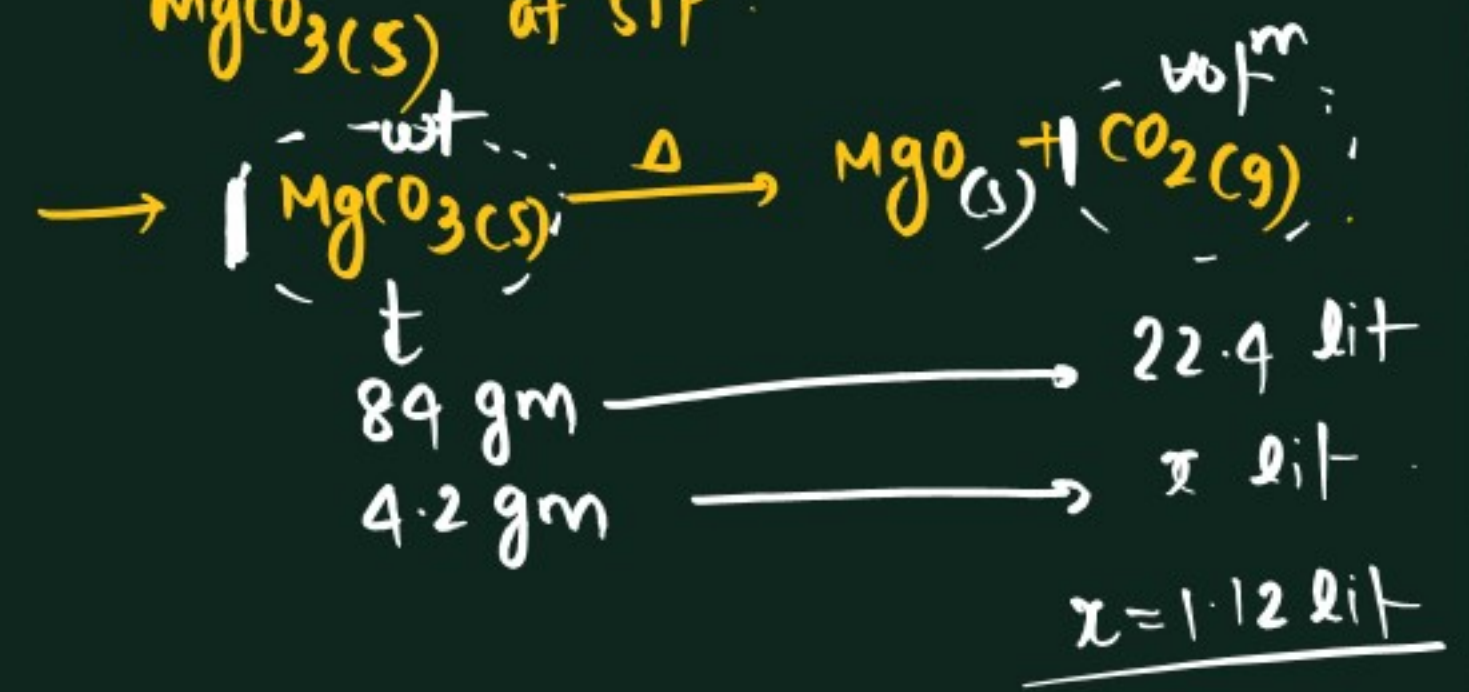
or.



$CaCO_3 = 10 \text{ gm}$
 Sand = 2 gm

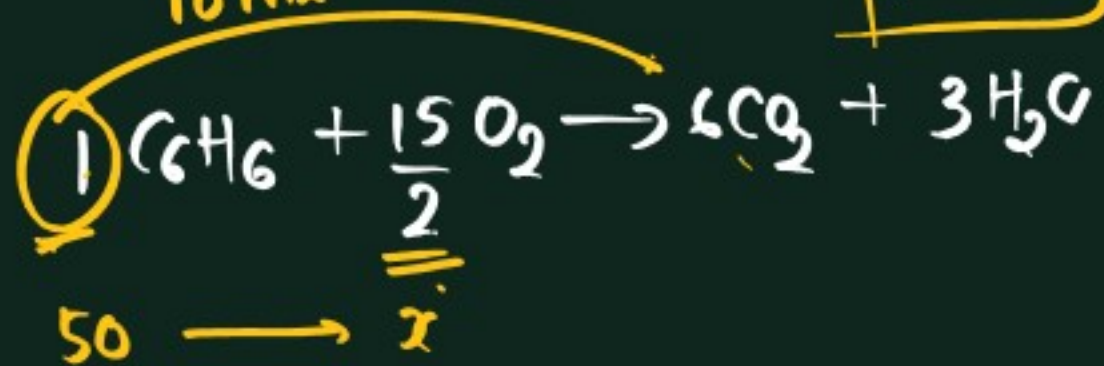
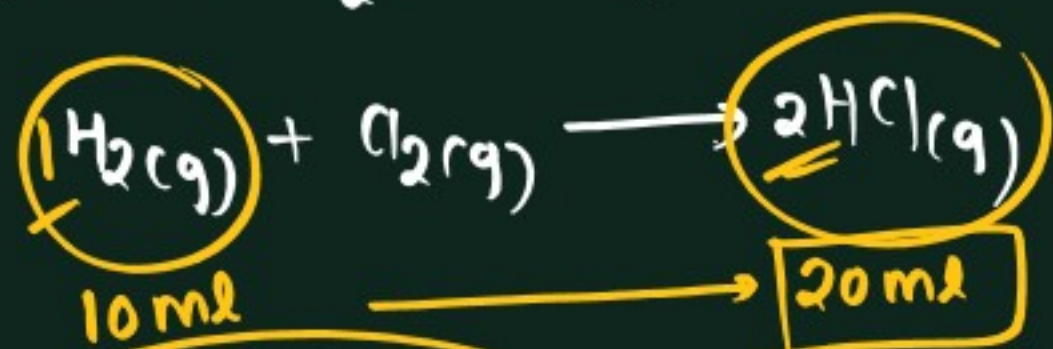
* wt-volume relation $\rightarrow$ (STP)

Q. 1) Calculate vol^m of $\text{CO}_2(\text{g})$ obtained by decomposition of 4.2 gm of $\text{MgCO}_3(\text{s})$ at STP.



* vol^m - vol^m relation →

- ① Calculate vol^m of HCl obtained from by reactⁿ of 10 ml of H₂ with Cl₂.



c) $V_{\text{O}_2} = \frac{21}{100} V_{\text{air}}$

$$V_{\text{air}} = \frac{100 \times V_{\text{O}_2}}{21} = \frac{100 \times 375}{21} = 1785 \text{ ml}$$

- ② for burning 50 ml of benzene calculate

a) vol^m of O₂ required for burning → 375 ml

b) vol^m of CO₂ produced → 300 ml

c) vol^m of H₂O produced → 150 ml

d) vol^m of air required that contains 21% by O₂ by vol^m

e) vol^m of H₂O(l) produced after condensation.

$$V = n \times 22.4$$

$$n = \frac{V(\text{ml})}{22.4}$$

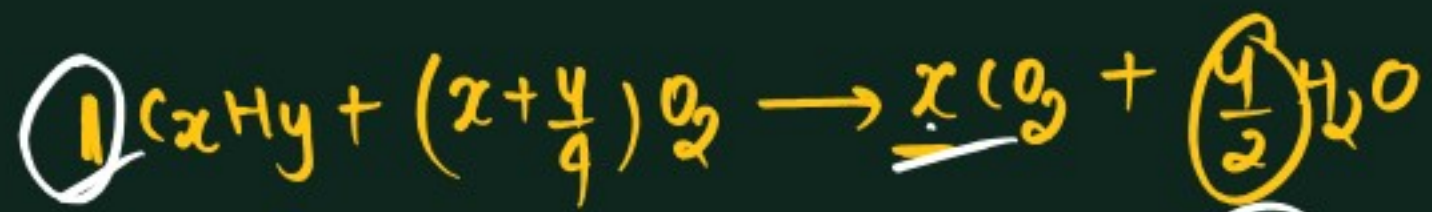
→ $V_{\text{H}_2\text{O}} = 150 \text{ ml}$

→ $n_{\text{H}_2\text{O}} = \frac{150}{22400} = \frac{15}{2240}$

→ $n_{\text{H}_2\text{O}} = \frac{15}{2240}$

→ $m_{\text{H}_2\text{O}} = \frac{15}{2240} \times 18$

→ $d = \frac{m}{V}$ $V = \frac{m}{d} = \frac{15 \times 18}{2240} \text{ ml}$



$\textcircled{10}$

40

$\textcircled{50}$

4

5

$x = 4$

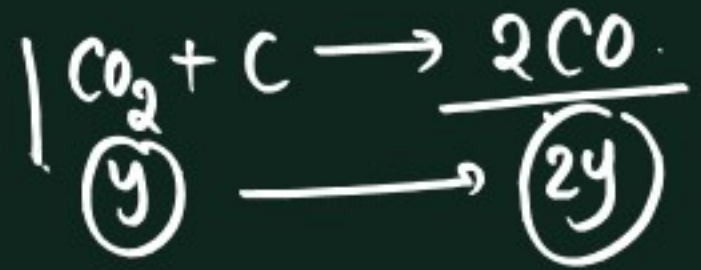
$\frac{y}{2} = 5$

$y = 10$

$\textcircled{C_4H_{10}}$



$$\boxed{\begin{matrix} CO = x \\ CO_2 = y \end{matrix}} \quad x + y = 100$$



$\begin{matrix} CO \\ CO_2 \end{matrix}$
100 ml



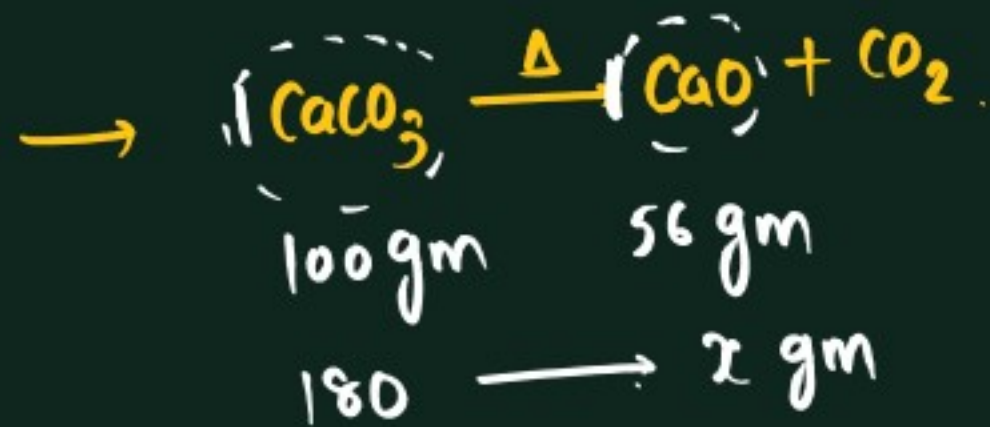
$\begin{matrix} \longrightarrow CO = x \text{ ml} \\ \longrightarrow CO = 2y \end{matrix}$
160 ml

$$\begin{matrix} x + 2y \\ = 160 \end{matrix}$$

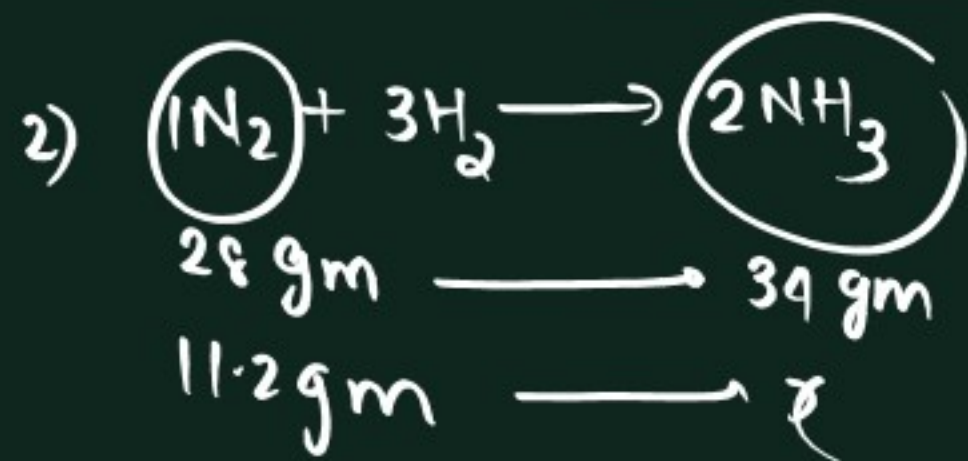
$y = 60 \text{ ml}$
 $x = 40 \text{ ml}$

∴ purity →

1) Calculate amt of CaO obtained by heating 200 gm of CaCO_3 which is 90% pure



$$x = 100.8 \text{ gm}$$



$$\begin{aligned} m_{\text{N}_2} &= 14 \times \frac{80}{100} \\ &= 11.2 \text{ gm} \end{aligned}$$

$$x = 13.6 \text{ gm}$$

$$\begin{aligned} m_{\text{CaCO}_3} &= \frac{90}{100} \times m_{\text{total}} \\ &= \frac{90}{100} \times 200 \\ &= 180 \text{ gm} \end{aligned}$$

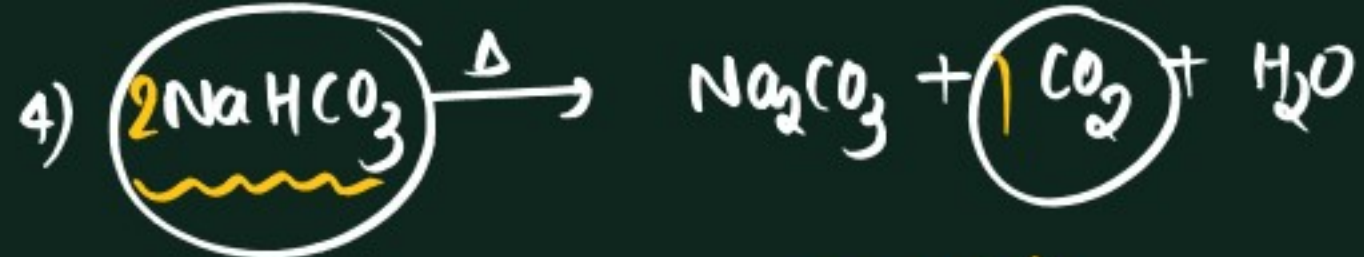


$$\begin{array}{l} 28 \text{ gm} \longrightarrow 1 \text{ mole} \\ 5.6 \longrightarrow n_{\text{O}_2} \end{array}$$

$$n_{\text{O}_2} = 0.2$$

$$\begin{aligned} \text{no. of 'o' atom} &= 2 \times n \times N_A \\ &= 2 \times 0.2 \times N_A \\ &= 0.4 N_A \end{aligned}$$

$$\begin{aligned} m_{\text{N}_2} &= 7 \times \frac{80}{100} \\ &= 5.6 \text{ gm} \end{aligned}$$



$168 \text{ gm} \longrightarrow 22.4 \text{ lit}$
 $x \longrightarrow 11.2 \text{ lit}$

$x = \text{NaHCO}_3 = 84 \text{ gm}$

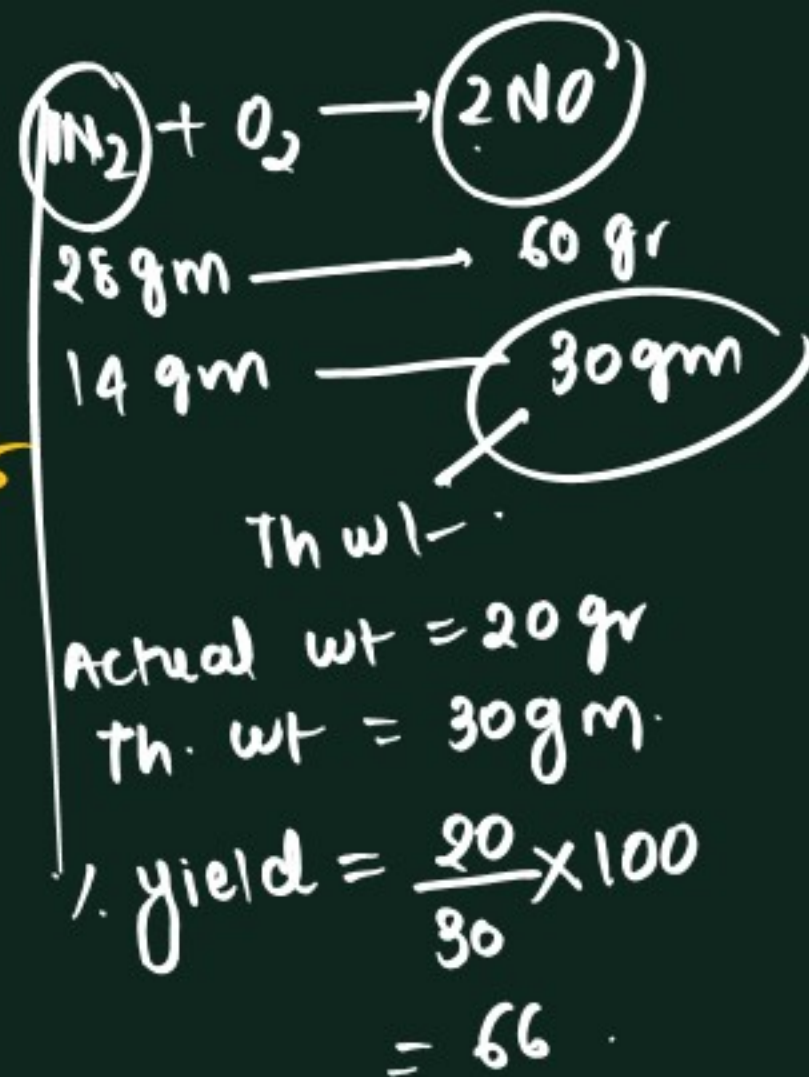
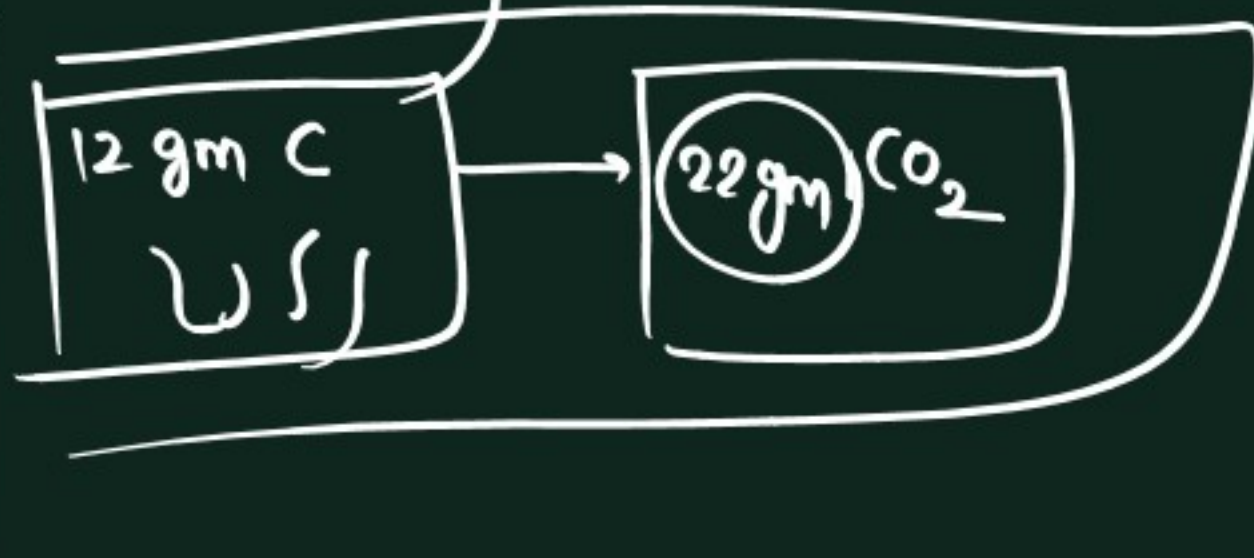
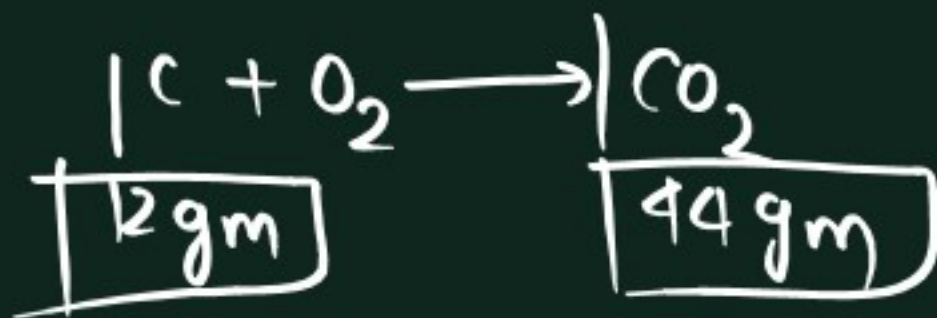
$m_{\text{NaHCO}_3} = \frac{80}{100} \times m_{\text{total}}$

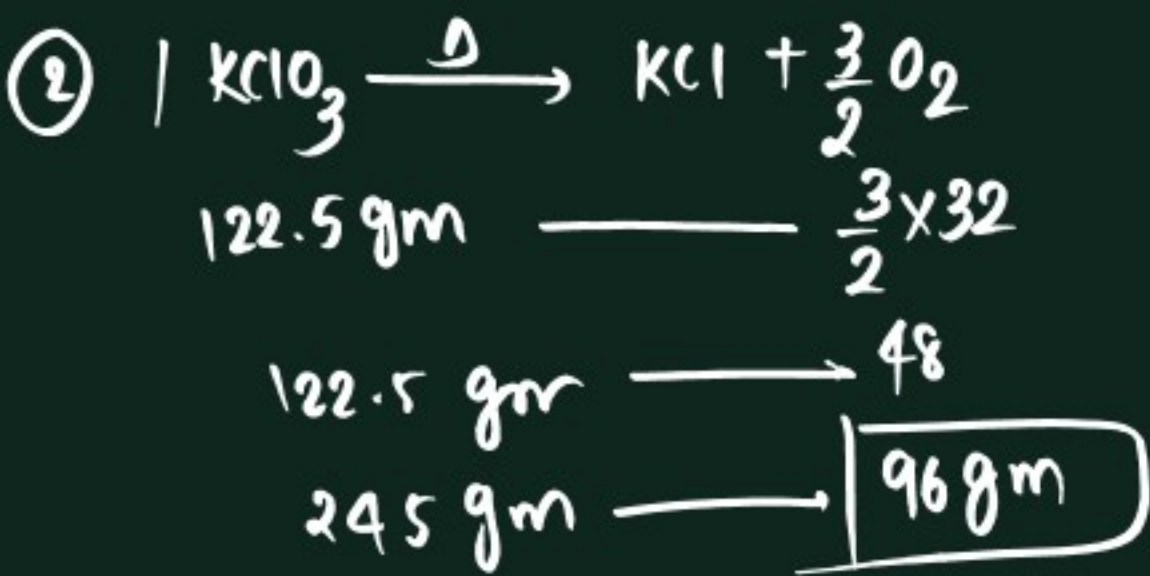
$84 = \frac{80}{100} \times m_{\text{total}}$

$m_{\text{total}} = \frac{100 \times 84}{80} = 105$

$\% \text{ yield} = \frac{\text{Actual wt. of product}}{\text{Theoretical wt. of product}} \times 100$
 $= \frac{22}{44} \times 100$

calculated from stoichiometry



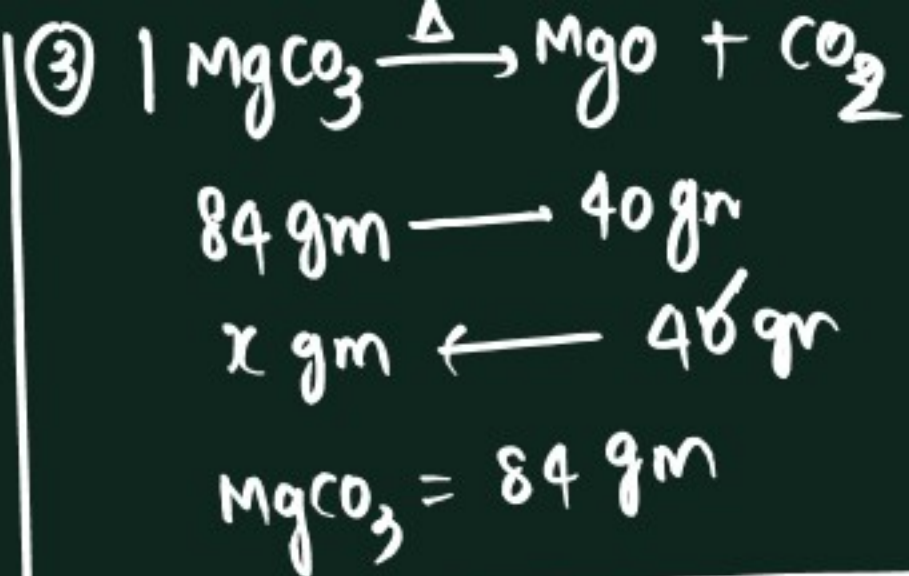


Th. wt = 96 gm

% yield = $\frac{\text{Actual wt}}{\text{Th. wt}} \times 100$

$31.25 = \frac{\text{Actual wt}}{96} \times 100$

Actual wt = 30 gm



$\text{MgCO}_3 = \frac{80}{100} \times m_{\text{total}}$

$84 = \frac{80}{100} \times m_{\text{total}}$

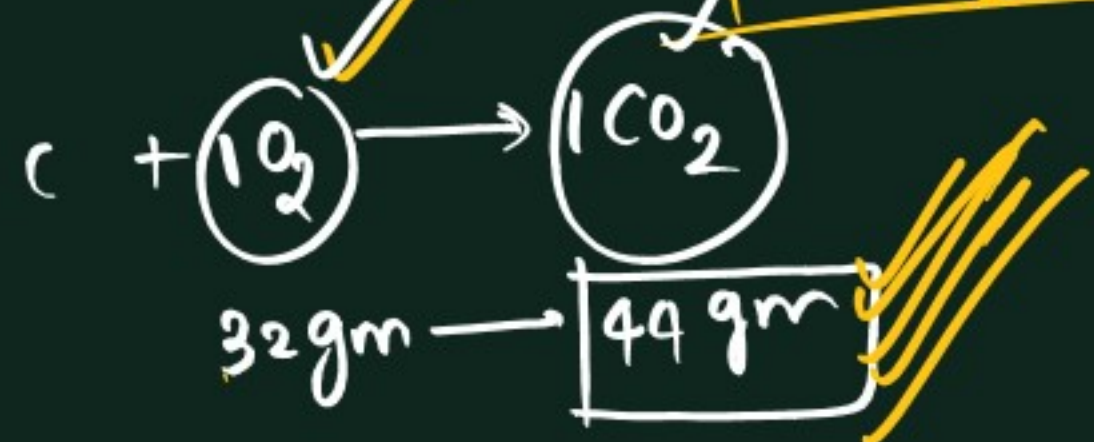
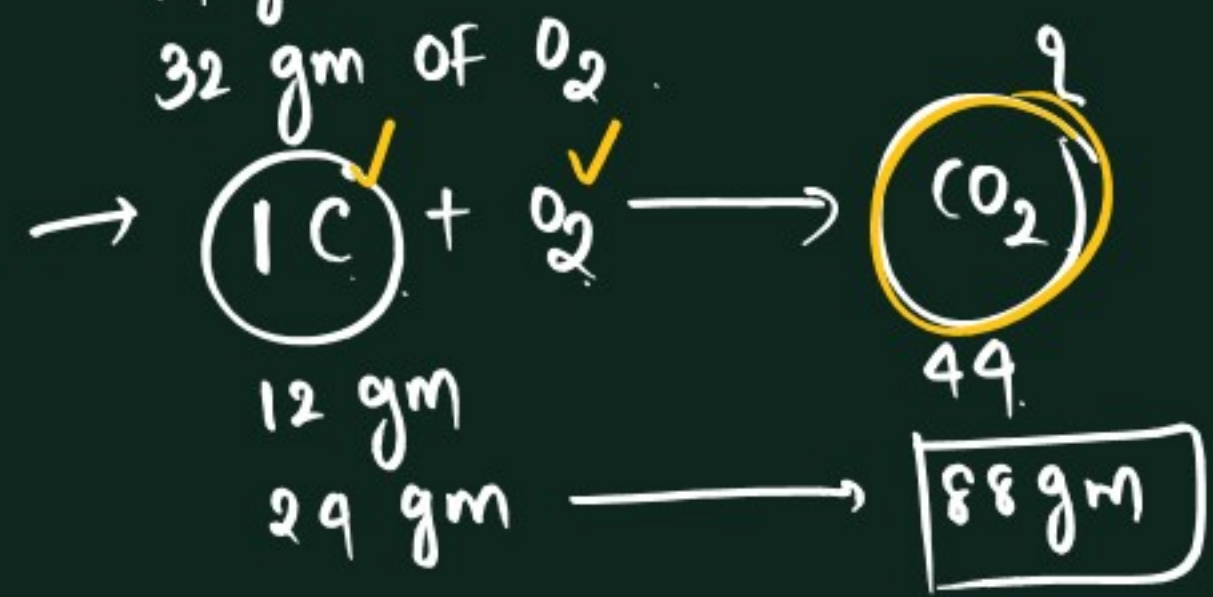
$m_{\text{total}} = 105 \text{ gm}$

% yield = $\frac{20}{\text{th wt}} \times 100$

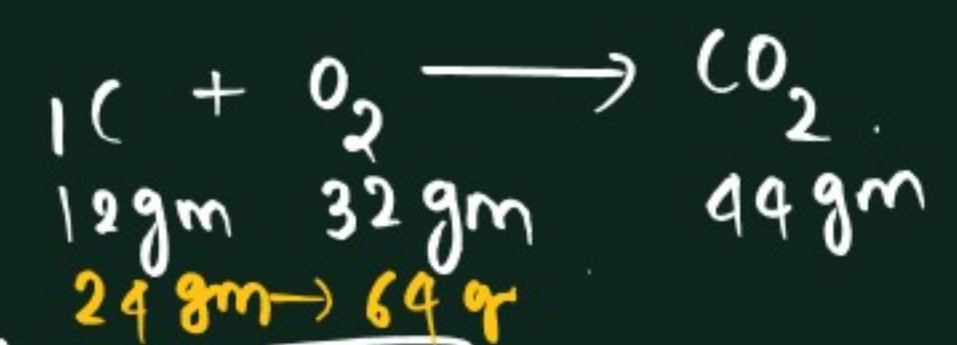
$50 = \frac{20}{\text{Th. wt}} \times 100$

Th. wt = 40 gm

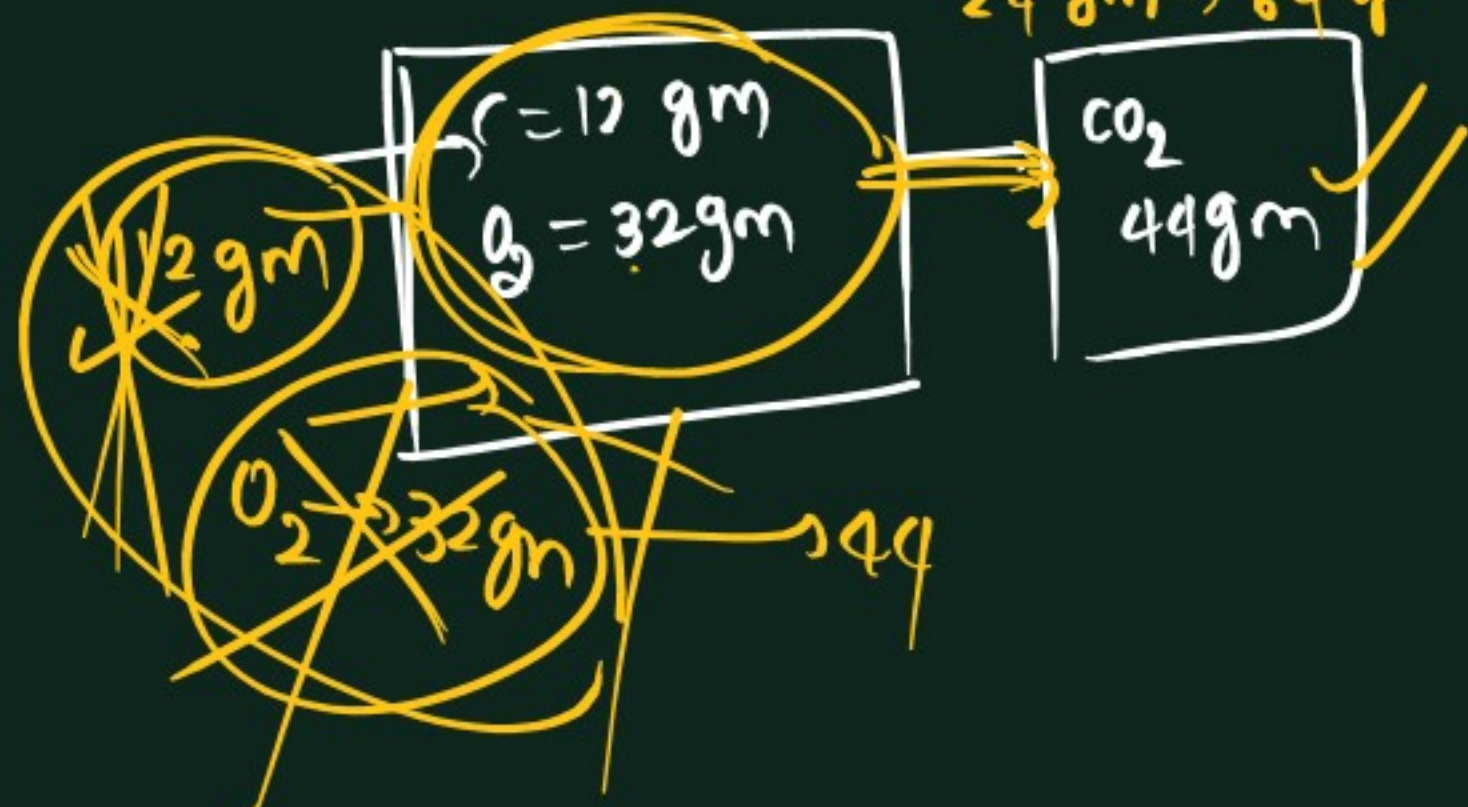
Q. Calculate amt. of CO_2 obtained by reaction of 24 gm of carbon with 32 gm of O_2



Limiting reagent.



$$\begin{array}{l} \text{C} \rightarrow \frac{n}{1} = \frac{2}{1} \\ \textcircled{\text{O}_2} \rightarrow \frac{n}{1} = \textcircled{\frac{1}{1}} \text{ L.R} \end{array}$$

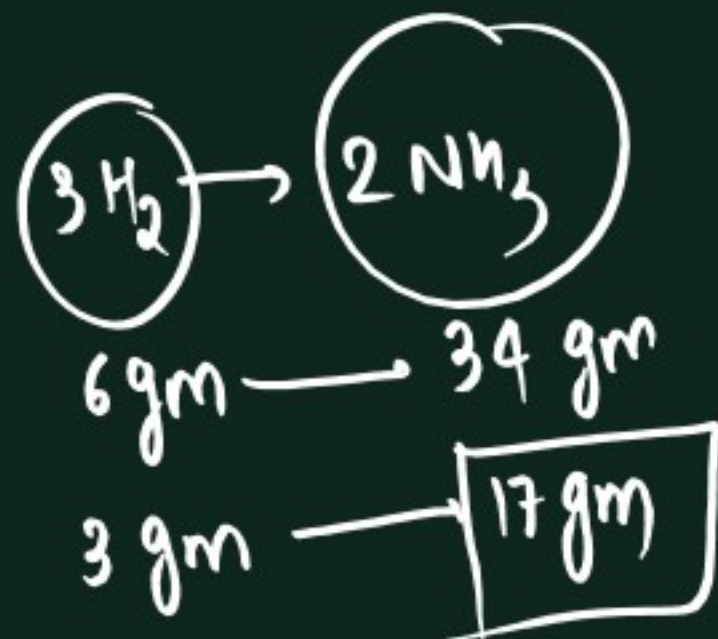


Limiting reagent $\rightarrow$
(L.R)

Excess reagent $\rightarrow$
(E.R)

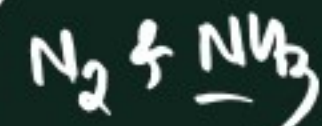
Determination of L.R $\rightarrow$

$$\text{Ratio} = \frac{\text{given moles}}{\text{S.C.}} \} \rightarrow \text{least} \rightarrow \text{L.R}$$



$$\text{NH}_3 = 17\text{ gm}$$

final mixt after rean $\rightarrow$



Q1)



$$\text{N}_2 = 28\text{ gm}$$

$$\text{H}_2 \rightarrow 3\text{ gm}$$

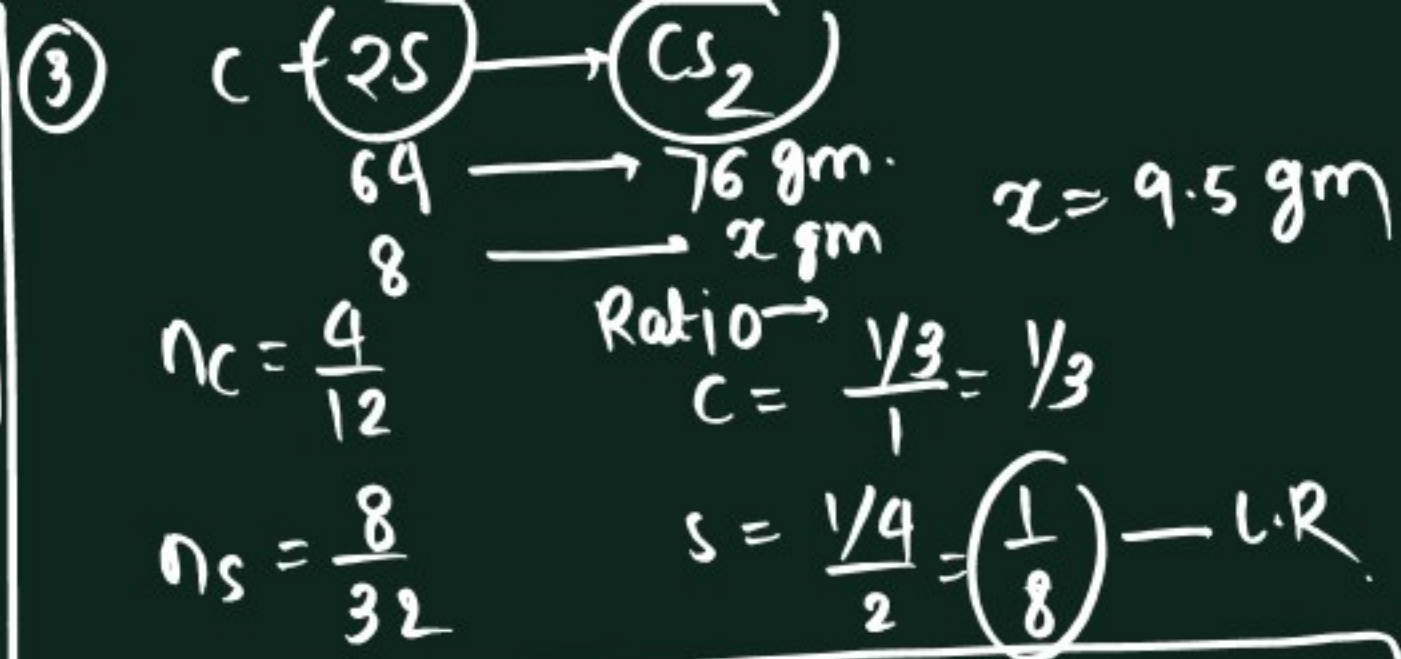
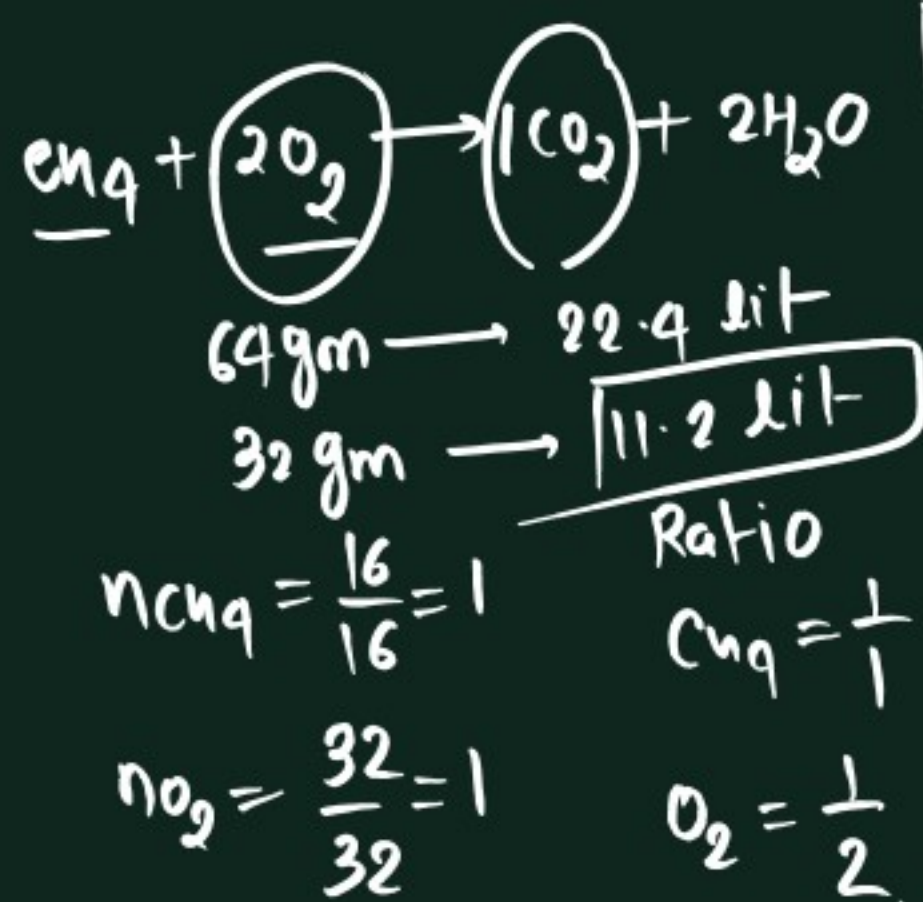
$$n_{\text{N}_2} = \frac{28}{28} = 1$$

$$n_{\text{H}_2} = \frac{3}{2} = 1.5$$

$$\text{Ratio} \rightarrow \text{N}_2 = \frac{1}{1}$$

$$\text{H}_2 = \frac{1.5}{3} = \frac{1}{2} \rightarrow \text{L.R}$$

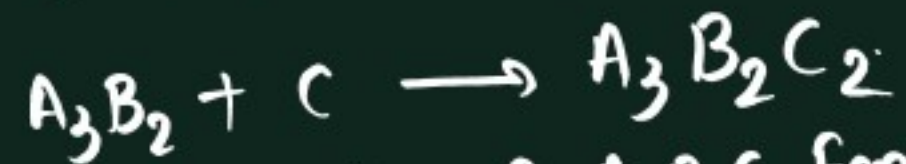
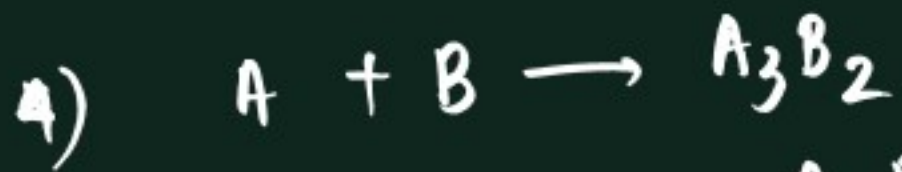
$$\begin{aligned} \% \text{ purity of product} &= \frac{\text{mass of product}}{\text{Total mass}} \times 100 \\ &= \frac{17}{31} \times 100 \end{aligned}$$



$$\% \text{ purity} = \frac{9.5}{12} \times 100 = 79\%$$

$$\text{amt of unreacted} = 12 - 9.5 = 2.5 \text{ gm}$$

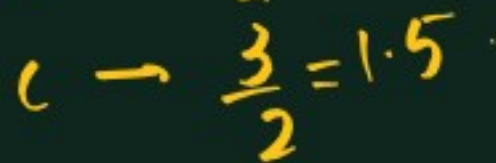
Yogesh



calculate moles of $A_3B_2C_2$ formed
if 3 moles of each A, B & C is taken



$A_3B_2C_2 = 1 \text{ mole}$



Principle of atom Conservation POAC

no. of atom of any element in reactant = no. of atoms of that element in product side



all carbon in C_3H_8 is converted into $\text{C}_3\text{H}_7\text{OH}$

POAC for C $\longrightarrow$

$$3 \times n_{\text{C}_3\text{H}_8} = 1 \times n_{\text{C}_3\text{H}_7\text{OH}}$$

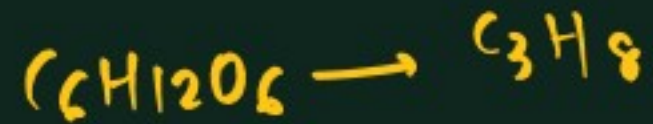
$$3 \times n_{\text{C}_3\text{H}_8} = n_{\text{C}_3\text{H}_7\text{OH}}$$



$$POAC \rightarrow C : 4 \times n_{C_4H_8} = 1 \times n_{CO_2}$$

$$POAC \rightarrow H \rightarrow 8 \times n_{C_4H_8} = 2 \times n_{H_2O}$$

$$POAC \rightarrow O : 2 \times n_{O_2} = 2 \times n_{CO_2} + 1 \times n_{H_2O}$$



$$6 \times n_{C_6H_{12}O_6} = 3 \times n_{C_3H_8}$$

$$\frac{6 \times 36}{180} = 3 \times \frac{V}{22.4}$$

$$V = 8.9 \text{ lit}$$



$$3 \times n_{Fe_3O_4} = 2 \times n_{Fe_2O_3}$$

$$3 \times \frac{\bar{w}}{Mw} = 2 \times \frac{x}{Mw}$$

$$3 \times \frac{5}{232} = 2 \times \frac{x}{160}$$

$$x = 5.17$$

$$Ex-1 \rightarrow 1-61$$

$$Ex-2 \rightarrow 1-27.$$

$$\% \text{ yield} = \frac{\text{actual vol}}{\text{Th. vol}^m}$$

actual vol^m =